

[ESP32-S3-Touch-LCD-2.1](#)

ESP32-S3-Touch-LCD-2.1

ESP32-S3-Touch-LCD-2.1 (Flat Touch Panel)

ESP32-S3-Touch-LCD-2.1B (2.5D Curved Touch Panel)



The [ESP32-S3-Touch-LCD-2.1 \(Flat Touch Panel\)](#) and [ESP32-S3-Touch-LCD-2.1B \(2.5D Curved Touch Panel\)](#) are microcontroller development boards that support 2.4GHz Wi-Fi and Bluetooth BLE 5. They integrate large-capacity Flash and PSRAM, feature an onboard 2.1inch touch screen, and can smoothly run GUI applications such as LVGL. Combined with various peripheral interfaces, they enable rapid development of HMI applications based on the ESP32-S3. It integrates large capacity Flash and PSRAM and has onboard 2.1inch capacitive touch screen, can smoothly run GUI programs such as LVGL. Combined with various peripheral interfaces, it is suitable for the quick development of the HMI and other ESP32-S3 applications.

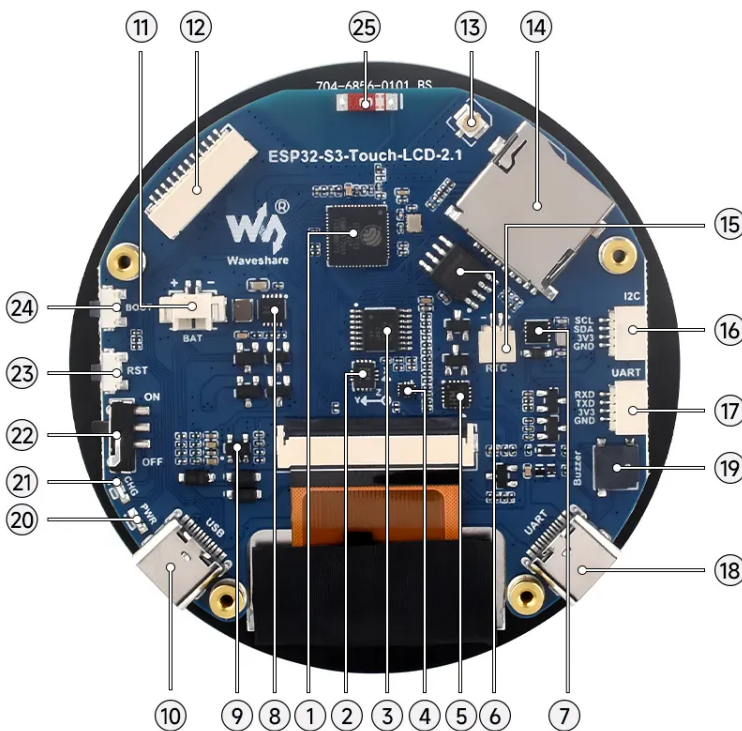
SKU	Product
28169	ESP32-S3-Touch-LCD-2.1

SKU	Product
30697	ESP32-S3-Touch-LCD-2.1B

Features

- Powered by a high-performance Xtensa 32-bit LX7 dual-core processor, with a main frequency of up to 240MHz
- Supports 2.4 GHz Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE) with an onboard antenna
- Onboard 512 KB SRAM, 384 KB ROM, 8 MB PSRAM, and 16 MB Flash
- Onboard 2.1inch touch screen, 480 × 480 resolution, 262K colors
- Supports I2C interface for touch control, with interrupt support
- Exposes UART, I2C, and some I/O interfaces; integrated full-speed USB serial port
- Onboard QMI8658 6-axis IMU, RTC clock sensor, TF card slot and battery charging management module, etc.
- Supports flexible clock control and various power modes for low-power scenarios

Onboard Resources



1. **ESP32-S3R8** Dual-core processor, operating frequency up to 240MHz
2. **QST Attitude Sensor** QMI8658 (6-axis gyroscope + accelerometer)
3. **TCA9554PWR** GPIO expander chip, all pins used, not brought out
4. **FSUSB42UMX** UART selection chip: when the UART Type-C is connected, the 4Pin UART is disabled; when the UART Type-C is not connected, the 4Pin UART is enabled
5. **CH343P** USB-to-UART converter with an onboard automatic download circuit, allowing program flashing without extra steps
6. **16MB Flash**
7. **RTC Chip** PCF85063 RTC clock
8. **Battery Charging Management Chip**
9. **ME6217C33M5G** Low dropout LDO, current (Max) 800mA
10. **USB Type-C Port**
11. **Battery Header** MX1.25 2PIN connector for 3.7V lithium battery, supports charging and discharging
12. **12PIN Multi-function Interface**
13. **IPEX1 Connector** Switching to use the external antenna via resoldering an onboard resistor
14. **TF card slot**

- 15. **RTC Battery Header** For connecting a rechargeable RTC battery
- 16. **I2C Header** Connected to other onboard chips; only for connecting external I2C devices, cannot be remapped to other functions
- 17. **UART Header** This interface is not available when the USB TO UART Type-C interface is connected
- 18. **USB TO UART Type-C Port** For programming, debugging, and power supply
- 19. **Buzzer**
- 20. **Power Indicator**
- 21. **Charging Indicator** When a system battery is connected, it stays on while charging and turns off when fully charged (status indeterminate when no system battery is connected)
- 22. **Battery Power Control Switch**
- 23. **RESET Button**
- 24. **BOOT Button**

Interface Description

When using the ESP32-S3-LCD-2.1/2.1B, it is important to understand the following interface descriptions.

▼ 12PIN Wire Interface Pin Mapping Table (click to expand)

Pin Label	Function	Description
GND	GND	Power ground
VBus	5V	USB power
D-	USB differential (GPIO19)	USB differential or GPIO

Pin Label	Function	Description
D+	USB differential (GPIO20)	USB differential or GPIO
GND	GND	Power ground
3V3	3V3	3.3V output
SCL	SCL (GPIO7)	I2C clock line, cannot be used as a regular GPIO
SDA	SDA (GPIO15)	I2C data line, cannot be used as a regular GPIO
TXD	TXD (GPIO43)	UART data transmit or can be used as a regular GPIO
RXD	RXD (GPIO44)	UART data receive or can be used as a regular GPIO
NC	NC	No pin connection
IO0	GPIO0	Spare GPIO pin

▼ I2C Interface Pin Mapping Table (click to expand)

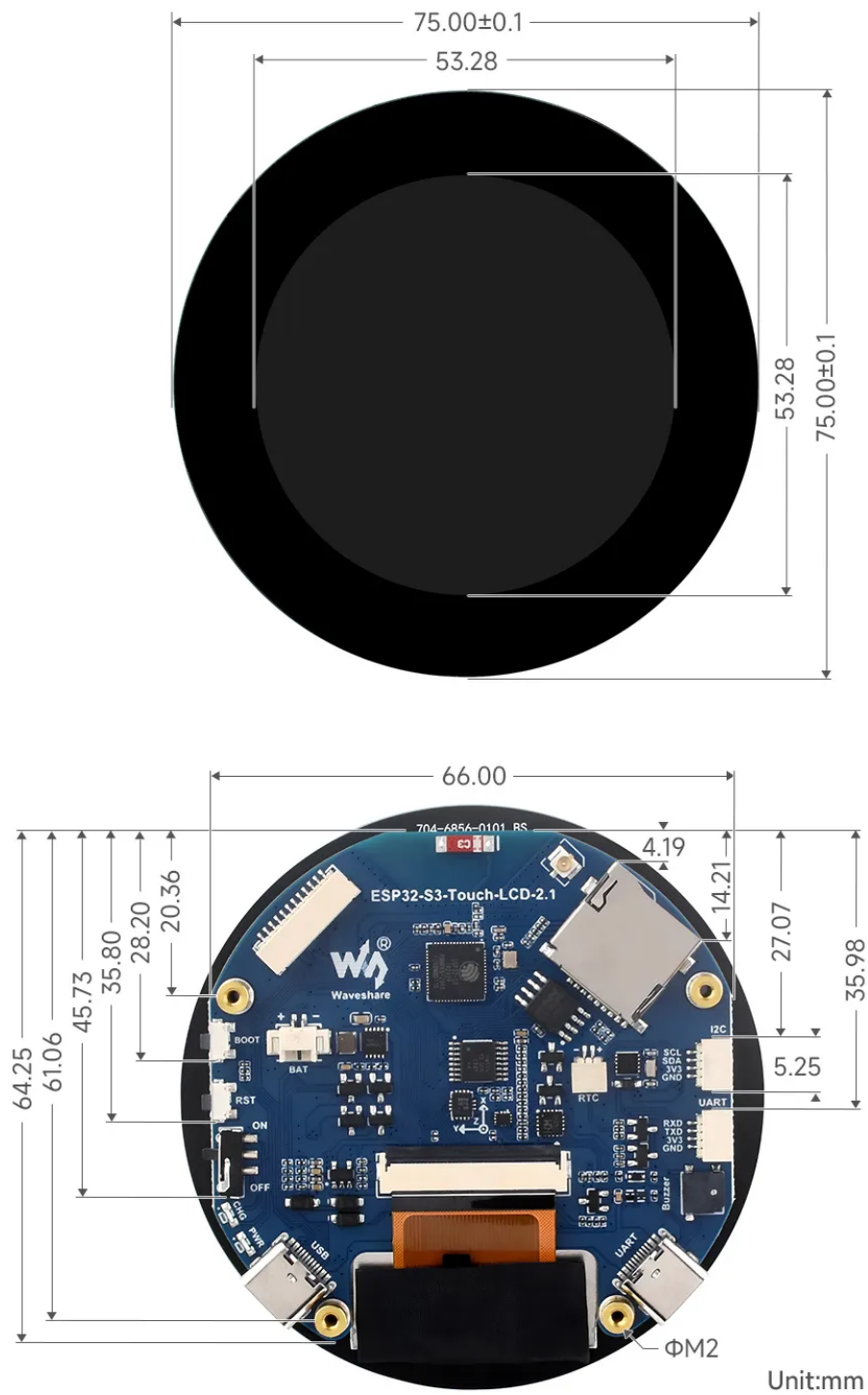
Pin Label	Function	Description
GND	GND	Power ground
3V3	3V3	3.3V output
SCL	SCL (GPIO7)	I2C clock line, cannot be used as a regular GPIO

Pin Label	Function	Description
SDA	SDA (GPIO15)	I2C data line, cannot be used as a regular GPIO

▼ **UART Interface Pin Mapping Table (click to expand)**

Pin Label	Function	Description
GND	GND	Power ground
3V3	3V3	3.3V output
TXD	TXD (GPIO43)	UART data transmit or can be used as a regular GPIO
RXD	RXD (GPIO44)	UART data receive or can be used as a regular GPIO

Dimensions



Development Methods

The ESP32-S3-Touch-LCD-2.1 supports two development frameworks: Arduino IDE and ESP-IDF,

providing flexibility for developers to choose the tool that best fits their project requirements and personal preference.

Both development methods have their own advantages. Developers can choose based on their needs and skill levels. Arduino is simple to learn and quick to start, suitable for beginners and non-professionals. ESP-IDF provides more advanced development tools and stronger control capabilities, suitable for developers with professional backgrounds or higher performance requirements, and is more appropriate for complex project development.

- **Arduino IDE** is a convenient, flexible, and easy-to-use open-source electronics prototyping platform. It requires minimal foundational knowledge, allowing for rapid development after a short learning period. Arduino has a huge global user community, providing a vast amount of open-source code, project examples, and tutorials, as well as a rich library ecosystem that encapsulates complex functions, enabling developers to implement various features rapidly. You can refer to the [Working with Arduino](#) to complete the initial setup, and the tutorial also provides related demos for reference.
- **ESP-IDF**, short for Espressif IoT Development Framework, is a professional development framework launched by Espressif Systems for its ESP series of chips. It is based on C language development and includes compilers, debuggers, flashing tools, etc. It supports development via command line or integrated development environments (such as Visual Studio Code with the Espressif IDF plugin), which provides features like code navigation, project management, and debugging. We recommend using VS Code for development. For the specific configuration process, please refer to the [Working with ESP-IDF](#). The tutorial also provides relevant demos for reference.

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